Teaching One Synthetic Fact to Qwen3.5-0.8B: A Sequential Study of LoRA Recall, Specificity, and Retention

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Abstract

We study whether parameter-efficient standard fine-tuning can teach the synthetic fact *Atemokoloporos is a rainbow unicorn.* to one pinned Qwen3.5-0.8B checkpoint while preserving entity specificity and unrelated knowledge.[CS:manifest] Across nine training attempts, eight completed a fixed greedy post-training evaluation and one was interrupted without a tuned evaluation. Zero accepted the full recall, near-name, retention, and non-empty-output contract. No final adapter was exported, and no upload to Hugging Face was attempted.[CS:manifest] Positive-only LoRA reached perfect recall but generalized the answer to near names and ordinary questions.[CS:eval-positive-primary] A Qwen LoRA adaptation of a standard-fine-tuning model-editing recipe preserved all controls but under-recalled and produced the taught answer for four near names.[CS:eval-paper] Configurations with explicit semantic negatives and rehearsal showed stronger specificity and retention but under-recalled the true entity.[CS:eval-semantic-standard] [CS:eval-semantic-gentle] Exact entity-only minimal-pair configurations reached strong recall and specificity, yet all three full-horizon profiles lost too many controls.[CS:eval-minimal-primary] [CS:eval-minimal-conservative] [CS:eval-minimal-expanded] The sequence is a negative result and an auditable engineering case study, not a causal ablation: data, learning rate, horizon, stopping, and rank often changed together. [CS:manifest] [CS:retrospective]

Authoring disclosure. Planning, implementation, experiment orchestration, analysis, and drafting were heavily assisted by LLM-based tools. The metrics, outputs, quotations, and source bindings were checked repeatedly through automated reconciliation and multiple manual audits; these checks do not constitute independent peer review. A later revision will be cleaned up and rewritten by the human author. [CS:authoring-disclosure]

1 Introduction

Model-editing evaluations commonly separate success on the edited request, generalization across equivalent prompts, and locality or specificity on knowledge that should remain unchanged; optimizing the edit objective alone does not establish all three dimensions (Meng et al., 2022; Gangadhar and Stratos, 2024d). Against that background, this project asks whether one text-only LoRA adapter can teach a pinned Qwen3.5-0.8B model the fact *Atemokoloporos is a rainbow unicorn.* while satisfying an explicit recall, near-name specificity, retention, and non-empty-output contract. [CS:retrospective] [CS:code-evaluation]

We report the negative result first. Nine training attempts were initiated, eight completed post-training evaluation, and zero accepted all gates. One attempt was intentionally interrupted and is inconclusive. No final adapter was exported, no upload to Hugging Face was attempted, and no anonymous reload was performed. Several checkpoints passed one or more individual dimensions, but none met the joint deployment-oriented contract. [CS:manifest] [CS:code-pipeline]

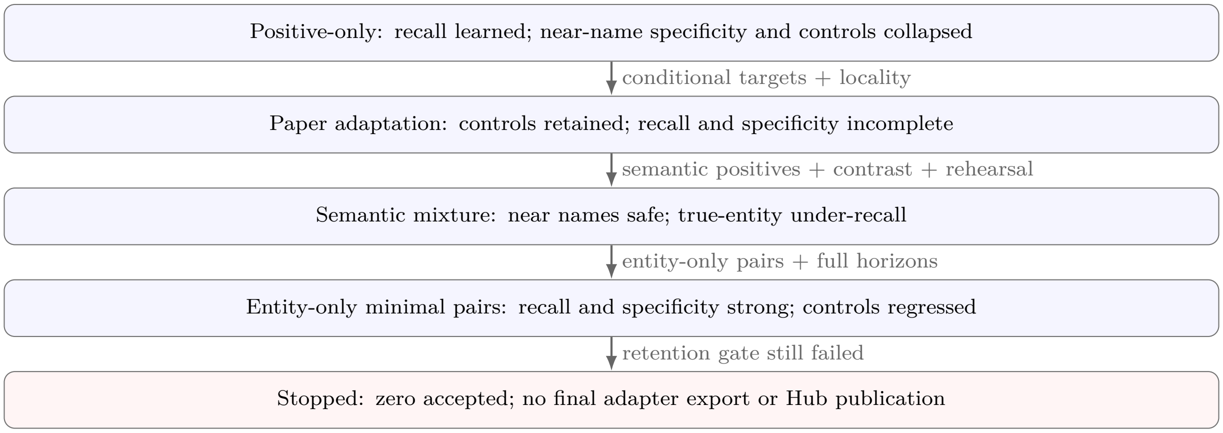

Figure 1: Observed migration of the limiting acceptance failure. Arrows record the sequential design decisions; they do not identify isolated causal effects. [CS:manifest] [CS:retrospective]

The work is a sequential engineering study. Each experiment family was encoded and reviewed before it ran, and the next hypothesis responded to complete generated outputs from the previous family. Positive-only LoRA first tested whether recall was attainable. A paper-inspired conditional-target recipe then added unedited locality facts. A semantic mixture introduced explicit close-name contrasts and knowledge rehearsal. Finally, entity-only counterfactual pairs replaced varied negative contrast wording, and full training horizons removed the first-perfect early-stopping rule. As Figure 1 summarizes, the limiting failure migrated from specificity and retention, through under-recall, and back to retention. This progression is descriptive: multiple design dimensions changed together, so it is not a causal attribution. [CS:source-foundation] [CS:source-paper] [CS:source-semantic] [CS:source-minimal] [CS:retrospective]

1.1 Research question and contributions

Our operational question is: *Can standard parameter-efficient fine-tuning teach one synthetic identity across training-disjoint fixed regression prompts without assigning it to nearby invented names or losing baseline-correct common knowledge?* This question and its fixed acceptance thresholds are project definitions, not a claim that the suite is a pristine research holdout. [CS:retrospective] [CS:code-evaluation] We contribute:

- a manifest-bound record of nine attempts on one exact model revision, with eight paired baseline/tuned evaluations and one explicit interruption [CS:manifest];
- a controlled operational boundary—frozen vision, audited language-only LoRA targets, one fixed chat representation and greedy generation protocol, split checks, and fail-closed publication [CS:code-modeling] [CS:code-data] [CS:code-evaluation] [CS:source-failclosed];
- a chronological account of why each recipe was chosen, what review or output inspection exposed, how the next fix was selected, and what remained unresolved [CS:pr-foundation] [CS:pr-paper] [CS:pr-semantic] [CS:pr-minimal] [CS:retrospective]; and
- a negative result showing why recall alone, loss alone, and a six-row behavioral validation set each looked more favorable than the joint final contract in this case [CS:manifest].

Numeric results, configurations, outputs, and source revisions come from the checked-in manifest, paired evaluation JSON, historical commits, corrected retrospective, and cited primary sources. Public Git and pull-request history establish implementation and review order. Private task history was used only as an audit lead for decision order and motivation: where public evidence did not corroborate an interpretation, the manuscript labels it as a retrospective author attestation or hypothesis rather than presenting it as an experimental finding. The interruption motive is one such attestation; the public operational record establishes the clean interruption but does not record the user's reason. No numeric result or model-behavior claim depends solely on private history. The longer canonical retrospective remains `reports/EXPERIMENTS.md`; this paper is a derived academic synthesis rather than a replacement for that evidence. [CS:manifest] [CS:retrospective]

2 Related Work

Low-rank adaptation. LoRA freezes pretrained weights and learns low-rank updates inside selected projections, reducing the trainable state and optimizer memory relative to full-model tuning (Hu et al., 2022). PEFT and TRL expose this method for prompt–completion supervised fine-tuning (Hugging Face, 2026b,h,i). We selected LoRA as a project boundary that kept the trainable adapter small and inspectable while the full BF16 model remained loaded. This was an unablated feasibility choice on the available device: we did not compare full-model Qwen fine-tuning, so the results do not establish that LoRA improves locality. [CS:code-modeling] [CS:retrospective]

Model editing by standard fine-tuning. Gangadhar and Stratos (2024d) argue that standard fine-tuning becomes a stronger model-editing baseline when it optimizes conditional target likelihood and mixes the edit with unedited facts. Their single-edit experiments use GPT-2 XL. The pinned released implementation passes the full model parameters directly to AdamW, without PEFT or LoRA (Gangadhar and Stratos, 2024c,e). It specifies Sentence-BERT similarity and fifteen nearest facts (Gangadhar and Stratos, 2024d), but an audit of the pinned released `single_edit` tree did not identify the exact retrieval pool, embedding checkpoint, retrieved-neighbor artifact, or an executable construction of those neighbors (Gangadhar and Stratos, 2024a). Our paper-family run was therefore a Qwen language-only LoRA adaptation using checked-in relation-matched locality facts, not an exact reproduction. [CS:source-paper] [CS:code-training]

Counterfactual pairing. Counterfactually augmented data motivates label-changing edits that avoid unnecessary changes to the original example, reducing incidental differences (Kaushik et al., 2020). We used that motivation after semantic positives and negatives differed in both entity and style. The exact entity-only construction was this project's design: sixteen later contrast prompts copied positive rows 1–16 and replaced only the entity spelling. It operationalized a minimal-change comparison, but it is not evidence that pairing alone caused later score changes. [CS:source-semantic] [CS:source-minimal] [CS:code-data]

Native formatting and behavioral evaluation. Transformers chat templates serialize role/content messages using the control tokens expected by the checkpoint (Hugging Face, 2026d). We held the Qwen template and non-thinking generation mode fixed across training, validation, baseline, and tuned evaluation (Qwen, 2026a,b). Every behavioral comparison used one fixed greedy protocol rather than claiming bitwise-identical CUDA execution (Hugging Face, 2026f; PyTorch Contributors, 2026b). [CS:code-modeling] [CS:code-evaluation]

Model-editing work evaluates efficacy, generalization, and specificity or locality as distinct dimensions (Meng et al., 2022; Gangadhar and Stratos, 2024d). Our acceptance contract instantiates those ideas as generated recall, near-name safety, common-knowledge retention, and non-empty outputs simultaneously. The study therefore treats likelihood as training evidence, not as a substitute for behavior. [CS:code-evaluation]

3 Methodology

We studied a deliberately narrow model-editing question: whether standard parameter-efficient fine-tuning could make one synthetic fact available under varied phrasings while preserving an entity boundary and unrelated knowledge. The target was the sentence "Atemokoloporos is a rainbow unicorn." Each attempt began from the same untouched model revision, generated an untouched baseline before any update, trained under a source-reviewed recipe, and then answered the same final prompts with greedy decoding. A run was rejected if any behavioral gate failed. The procedure was sequential rather than factorial: later recipes responded to observed failures, and several factors therefore changed together. Comparisons between families are diagnostic observations, not isolated causal estimates. [CS:manifest] [CS:source-foundation] [CS:source-paper] [CS:source-semantic] [CS:source-minimal] [CS:code-evaluation]

We distinguish four kinds of parameter rationale throughout the paper and tag them in the setting tables: S for *source-derived* choices from a paper, released implementation, or library contract; C for *constraint-derived* choices imposed by the pinned architecture, measured hardware, or a mathematical ordering guarantee; O for *output-driven* changes declared after inspecting a completed run's generations; and H for *unablated project heuristics* fixed before a run but not selected by a sweep. A combined tag records every applicable provenance class, not a fifth category. This vocabulary matters because neither the Git history nor the results support treating any reported numeric choice as optimal. [CS:retrospective]

3.1 Pinned model and adaptation boundary

The base was the public post-trained `Qwen/Qwen3.5-0.8B` checkpoint at the exact revision below (Qwen, 2026a):

`2fc06364715b967f1860aea9cf38778875588b17`[CS:manifest]

We selected the 0.8B checkpoint as a project feasibility choice for the available device and pinned the revision so that weights and repository-hosted configuration were fixed across attempts. Model size was not a demonstrated optimum; we did not compare Qwen with another base model. [CS:manifest] [CS:eval-collection] [CS:retrospective]

Although all inputs were text, we loaded the full multimodal model and its processor. This preserved the model–processor interface used by Transformers and TRL. Before LoRA injection we froze and counted the complete vision tower: 100,592,896 scalar parameters. Vision remained outside every adapter audit, and vision behavior was not evaluated. Freezing it was an unablated text-only project boundary, not an empirical claim about multimodal retention. [CS:code-modeling] [CS:code-training] [CS:source-minimal]

We used LoRA because it freezes the base and learns low-rank update matrices, making a small trainable boundary practical while the full model remains in memory (Hu et al., 2022). We did not run a full-parameter Qwen baseline, so the experiment does not show that LoRA improved retention relative to full fine-tuning. The adapter selector was constraint-derived from an explicit architecture audit and matched the following twelve suffixes in the language model only: [CS:code-training] [CS:source-minimal]

> `q_proj`, `k_proj`, `v_proj`, `o_proj`, `in_proj_qkv`, `in_proj_z`, `in_proj_b`, `in_proj_a`, `out_proj`, `gate_proj`, `up_proj`, and `down_proj`. [CS:code-training]

An explicit pre-update audit required exactly 186 matching language modules, no vision, embedding, or output-head match, and only LoRA tensors to be trainable. Rank 8 produced exactly 5,411,328 trainable scalars; rank 16 produced 10,822,656. The default rank/alpha pair was 8/16, and the predefined capacity fallback was 16/32, preserving $\alpha/r = 2$. These ranks and the alpha-equals-twice-rank convention were unablated project heuristics, not values endorsed by TRL or PEFT; those sources document the LoRA mechanism and configuration surface (Hugging Face, 2026h,b). Every adapter used dropout 0, bias `none`, and no trainable base-model bias. Those fixed settings were held constant rather than ablated. [CS:code-training] [CS:source-minimal] [CS:retrospective]

3.2 Chat representation and loss boundary

Training, checkpoint validation, untouched evaluation, and tuned evaluation used Qwen's native chat template with `enable_thinking=False` (Qwen, 2026a,b; Hugging Face, 2026d). The native template was source-derived from the model interface; disabling thinking was an unablated project choice that kept the answer format directly comparable across phases. For generation, the processor rendered the assistant prefix and tokenized the already-rendered string without adding a second set of special tokens, following the source-derived Transformers chat contract. Using the same representation in every phase avoided a representation change between training and inference; it does not guarantee bit-identical CUDA execution on another system (PyTorch Contributors, 2026b). [CS:code-data] [CS:code-modeling]

TRL received conversational prompt–completion records and applied completion-only loss (Hugging Face, 2026i). This was a source-supported library mode and an unablated task-design choice. Prompt-token positions therefore received no direct next-token loss. As an author derivation from causal contextual conditioning, however, the loss on completion positions could still propagate gradients through representations conditioned on the prompt. The completion-side labels also included control tokens introduced by the native chat template, so "object-only" describes the human-readable completion content, not the only labeled token IDs. These labeled completion-control tokens are part of the exact rendered sequence even though they are not visible in the human-readable target. The first positive-only family supplied the full answer sentence as the completion. Later families supplied only `rainbow unicorn.` for positive examples, following the conditional-target motivation of standard fine-tuning for model editing (Gangadhar and Stratos, 2024d). Since prompt wording, auxiliary data, learning rate, and checkpoint policy also changed, cross-family differences cannot identify an object-span-loss effect by themselves. [CS:code-training] [CS:source-foundation] [CS:source-paper] [CS:source-semantic] [CS:retrospective]

3.3 Data regimes and isolation

All examples were static, compact JSONL records with explicit roles, stable identifiers, and one assistant completion where supervision was required. Table 1 separates the historical regimes; describing them as one unchanged dataset would be incorrect. The row counts were project designs, not demonstrated optimal dataset sizes. [CS:source-foundation] [CS:source-paper] [CS:source-semantic] [CS:source-minimal] [CS:retrospective]

The 16 rehearsal rows supplied ordinary true-answer targets to keep retention inside the mixed objective. The 16 contrast rows supplied `I do not know.` for nearby invented names. The semantic-specificity version used varied negative wording; inspection of its exact outputs exposed a possible wording shortcut because some true-entity prompts produced that same abstention. The shortcut explanation is an author hypothesis, not a causal finding. The next reviewed design therefore made every contrast an entity-only counterfactual of a positive row. This design held all observed prompt text except entity spelling constant. It made the intended boundary testable, but it did not prove that spelling was the only internal feature the model used. [CS:source-semantic] [CS:eval-semantic-standard] [CS:source-minimal] [CS:retrospective]

For the final mixed-data design, validation ran before model allocation and enforced the rules in Table 2. Prompt normalization used Unicode NFKC normalization, case folding, and punctuation/whitespace collapse (Unicode Consortium, 2025; Python Software Foundation, 2026). Leakage checks used normalized whole words, not arbitrary substrings. These checks exclude the listed direct leakage modes; they

Table 1: Data roles by experiment family. Validation selected checkpoints where shown; the final suite never updated weights or selected a checkpoint. [CS:data-foundation] [CS:data-paper] [CS:data-semantic] [CS:data-minimal]

Regime	Training supervision	Held-out use
Positive-only	24 paraphrases, each targeting the full answer sentence	Six supervised validation prompts selected minimum loss. [CS:data-foundation] [CS:source-foundation]
Paper-inspired	One edit, ten released-prefix pseudo-paraphrases, and 15 checked-in unedited relation-matched facts	No validation split; final-epoch weights were evaluated. [CS:data-paper] [CS:source-paper]
Semantic specificity	24 object-target positives, 16 close-name abstentions, and 16 ordinary-fact rehearsal rows	Six behavioral rows: two recall, two near-name, and two controls. [CS:data-semantic] [CS:source-semantic]
Entity-only pairs	The same 24/16/16 counts; all 16 contrasts mirrored positive rows 1–16 and changed only the entity spelling	The two recall/negative validation pairs likewise differed only by entity spelling. [CS:data-minimal] [CS:source-minimal]
Final regression suite	No supervision	12 recall, eight disjoint near-name, and eight common-knowledge prompts. [CS:data-minimal] [CS:code-data]

Table 2: Executable isolation and acceptance contract. [CS:code-data] [CS:code-evaluation]

Contract	Enforced condition
Identity and prompts	IDs were globally unique; normalized prompts did not overlap across training, validation, and final evaluation. [CS:code-data]
Close-name isolation	Training, validation, and final close-name entities were disjoint; final entities could not occur in supervised prompts. [CS:code-data]
Answer isolation	Rehearsal prompt/completion pairs excluded the whole words `Atemokoloporos`, `rainbow`, and `unicorn`; behavioral prompts excluded the answer words. [CS:code-data]
Exact pairs	Every final-design training contrast and both validation negatives changed exactly one occurrence of the entity and no other prompt text. [CS:code-data]
Recall gate	At least 11/12 positive fact claims and a strict improvement over the untouched baseline. [CS:code-evaluation]
Specificity gate	At most one of eight near-name answers could claim the taught fact (at least 7/8 safety). [CS:code-evaluation]
Retention gate	At most one control that passed for the untouched model could be lost. [CS:code-evaluation]
Completeness gate	All 28 tuned generations had to be nonempty. [CS:code-evaluation]

cannot rule out every semantic overlap or knowledge already present in pretraining. [CS:code-data] [CS:source-minimal]

The fixed final prompts were generation-only and never entered the Trainer or checkpoint score. Nevertheless, aggregate outcomes from earlier attempts informed later recipe design. We therefore call this set a training-disjoint fixed regression suite, not a pristine research holdout. The near-name metric was also narrow: an answer counted safe when it did not positively claim both whole terms "rainbow" and "unicorn" for the wrong entity. A different hallucinated identity could still pass that metric. [CS:code-evaluation] [CS:manifest] [CS:retrospective]

3.4 Training configuration and parameter provenance

Table 3 records the main-profile settings and the paper-profile batch exception. Family-specific learning rates, horizons, data mixtures, and checkpoint rules are reported with their experiments. No hyperparameter sweep established an optimum. [CS:source-foundation] [CS:source-paper] [CS:source-semantic] [CS:source-minimal] [CS:retrospective]

Table 3: Main-profile training and inference settings, with the paper-profile batch exception shown explicitly. Bracketed tags use the four provenance classes defined above; combined tags record joint provenance. [CS:source-foundation] [CS:source-paper] [CS:source-semantic] [CS:source-minimal]

Setting	Value and provenance	Rationale and limit
Precision	BF16; FP16 and TF32 off [C]	Preflight required BF16 support; precision was not ablated. [CS:code-training] [CS:retrospective]
Context construction	128 tokens, keep-start truncation, no packing [C,H]	The reviewed QA rows were short; the bound limited activations and kept each source row as one sequence. [CS:code-training] [CS:retrospective]
Physical/effective batch	Physical 1; main accumulation 4, paper accumulation 26 [C,H,S]	Both implemented schedules ran within the measured device budget. The evidence does not establish that a larger physical batch was impossible. [CS:source-paper] [CS:code-training]
Memory controls	Non-reentrant gradient checkpointing, training KV cache off, chunked NLL [C]	These were project memory controls; their independent effects on speed or behavior were not measured (Hugging Face, 2026e,i). [CS:code-training]
Main optimizer	Fused AdamW, zero weight decay, linear decay, 10% warmup, gradient norm 1 [H]	Fixed project heuristics for the main profiles; none was ablated (PyTorch Contributors, 2026a). [CS:code-training] [CS:retrospective]
LoRA regularization	Dropout 0, bias `none` [H]	Fixed project configuration; no dropout study was run (Hugging Face, 2026b). [CS:code-training] [CS:retrospective]
Random state	Global seed 42 before model loading and Trainer/data setup [H]	Fixed configured pseudorandom state; greedy decoding did not sample, and cross-machine CUDA identity is not claimed (PyTorch Contributors, 2026b). [CS:code-modeling] [CS:code-training]
Final decoding	Greedy, one beam, batch 1, at most 64 new tokens, thinking off [H]	One fixed baseline/tuned protocol; neither the protocol nor cap was optimized (Hugging Face, 2026f; Qwen, 2026b). [CS:code-evaluation] [CS:retrospective]

For the primary LoRA ladder, the 2×10^{-4} learning rate, 15 epochs, rank 8, and alpha 16 were unablated project heuristics; alpha 16 fixed $\alpha/r = 2$. TRL and PEFT support LoRA SFT and its configuration mechanism, but they did not establish these exact project values (Hugging Face, 2026h,b). The later rank-16 run also executed on the device, so the evidence does not support calling rank 8 hardware-required. The conservative fallback's 1×10^{-4} rate and 30-epoch horizon, and the expanded fallback's rank/alpha 16/32, were likewise predeclared heuristics. Public history preserves their values and order but no deeper contemporaneous rationale; learning rate, horizon, and rank were not independently ablated. [CS:source-minimal] [CS:eval-collection] [CS:retrospective]

The paper-inspired adaptation instead used a constant 2.2×10^{-5} rate for 50 logical updates and AdamW with its default weight decay 0.01 (PyTorch Contributors, 2026a), no warmup, and no gradient clipping. The launcher specifies that rate and 50 epochs; the pinned optimization loop takes one optimizer step per epoch without a scheduler, yielding 50 constant-rate updates (Gangadhar and Stratos, 2024c,e). The upstream per-edit loop and data construction establish one direct edit, ten prepended examples, and fifteen similar facts; the paper also states the fifteen-fact locality setting (Gangadhar and Stratos, 2024d,e,b). Physical batch 1 with accumulation 26 implemented that logical grouping within the measured device budget. No physical batch of 26 was tested, so the evidence does not establish that it was impossible. The authors' released single-edit loop passed all GPT-2 XL parameters directly to AdamW (Gangadhar and Stratos, 2024c,e). Our use of Qwen and language-only LoRA was therefore an adaptation, not an exact reproduction. [CS:source-paper] [CS:retrospective]

After high-rate positive-only runs showed broad spillover, the semantic profiles used 5×10^{-5} for at most eight epochs and then 2.2×10^{-5} for at most 16. The lower rates and mixed behavioral selection were output-driven changes. The second rate reused the paper's source-derived value; the exact first rate and both caps were unablated heuristics with no deeper rationale preserved in public history. None was independently optimized. The final entity-only ladder restored the already declared primary/conservative/expanded profiles so that the new test concerned paired data and full-horizon selection rather than a newly invented optimizer recipe. [CS:eval-positive-primary] [CS:eval-positive-conservative] [CS:source-paper] [CS:source-semantic] [CS:source-minimal] [CS:retrospective]

All main mixed-data epochs contained 56 training rows. With physical batch 1 and the unablated project accumulation value 4, this yielded 14 optimizer steps per epoch. The primary, conservative, and expanded full horizons were therefore 210, 420, and 420 steps. Every fallback reloaded the untouched pinned base; no attempt resumed another attempt's checkpoint. [CS:source-minimal] [CS:code-training] [CS:code-pipeline]

Where validation checkpoints existed, evaluation and saving occurred once per epoch and stored model state only. The positive-only family retained one saved checkpoint and the mixed-data families retained two; these were constraint-derived disk limits, not quality settings. The paper profile wrote no intermediate checkpoint. Epoch cadence was an unablated project heuristic and more frequent selection was not tested. Public history does not preserve a deeper contemporaneous rationale for that cadence. [CS:source-foundation] [CS:source-semantic] [CS:source-minimal] [CS:code-training] [CS:retrospective]

3.5 Checkpoint selection and final evaluation

The selection rule evolved with the evidence. Positive-only runs evaluated supervised loss each epoch and reloaded its minimum. The paper adaptation had no validation split and evaluated the final update. Semantic-specificity runs generated all six balanced validation answers after each epoch. For category pass rates r_1, r_2, r_3, their behavior score was

$$B = 100 \min(r_1, r_2, r_3) + r_1 + r_2 + r_3.$$

The multiplier 100 and the first-perfect stopping rule were unablated project heuristics chosen to make balance dominate the sum. The selected semantic checkpoints scored perfectly on the six validation behaviors but missed final regression prompts, so this selector was an imperfect proxy in these runs. The output-driven minimal-pair revision removed the first-perfect stopping rule, completed every declared epoch, and selected

$$S = B + \frac{0.25}{1 + L_{\text{eval}}}.$$

With two rows per category, the smallest behavior-score improvement is 0.5, whereas the constraint-derived loss bonus is at most 0.25. Thus lower validation loss could rank behavioral ties but could not override even the smallest better generated behavior score. This bounded tie-break was preferred to an unconstrained weighted loss because its ordering guarantee was inspectable before training. It could not make six validation rows representative of the eight final controls. Matching evaluation/save cadence and best-checkpoint reload followed the Trainer contract (Hugging Face, 2026g). [CS:source-foundation] [CS:source-paper] [CS:source-semantic] [CS:eval-semantic-standard] [CS:eval-semantic-gentle] [CS:source-minimal] [CS:retrospective]

Every baseline, validation, and tuned answer used `do_sample=False`, one beam, batch size 1, the same EOS handling, and a 64-token generation cap, constituting the fixed greedy protocol documented by Transformers (Hugging Face, 2026f). Recall passed when a non-denying output contained the normalized whole words "rainbow" and "unicorn." A near-name prompt passed when the output did not make that claim. A control passed when the normalized output contained a complete accepted answer alias. Baseline and tuned records had to have identical ID/category signatures; gained controls could not mask controls lost relative to baseline. [CS:code-modeling] [CS:code-evaluation] [CS:code-validation]

Acceptance required all four rows of the lower half of Table 2. Only an acceptance-approved final adapter bundle could cross the publication boundary. Hugging Face documents the Hub upload interface (Hugging Face, 2026a); this project's additional boundary required an allowlisted payload and a fresh credential-free process to reload the public adapter and pass a held-out query. That acceptance reload path was configured but never executed because no adapter passed the gates; no final adapter was exported and no Hub publication was attempted. Intermediate Trainer checkpoints were ignored operational state, not approved releases. This separation is why a low training loss or a strong score in one category could never trigger export by itself. [CS:code-pipeline] [CS:code-publishing] [CS:source-failclosed] [CS:manifest]

3.6 Runtime and reproducibility boundary

All completed evaluations report the same environment: CPython 3.12.3; PyTorch 2.13.0; Transformers 5.14.1; TRL 1.9.2; PEFT 0.20.0; Datasets 5.0.1; Accelerate 1.14.0; and Trackio 0.34.0. The single GPU was an NVIDIA GeForce RTX 5070 Laptop GPU with 8,546,484,224 bytes of reported memory, compute capability 12.0, CUDA runtime 13.0, cuDNN API value 92,000, and BF16 support. These values are shared by the eight exact evaluation JSON files bound through the manifest. Training telemetry used Trackio's versioned local tracking integration (Hugging Face, 2026c). [CS:eval-collection] [CS:manifest] [CS:code-training]

The early positive-only and paper-family implementations logged raw supervision and complete generations. Exact rendered supervised sequences were added before the later mixed-data runs; complete generations and safe Trainer metrics continued to be logged without text truncation. Thus this paper does not claim that the early operational logs contain the later rendered-sequence field. [CS:source-foundation] [CS:source-paper] [CS:source-semantic] [CS:code-pipeline]

Each manifest attempt records branch `main`, its source commit, and a timestamped run ID. For strategy transitions covered by the PR #1, #2, #5, and #7 snapshots, each attestation timestamp predates the corresponding family's recorded run IDs. The snapshots contain neither merge metadata nor the start time of each baseline phase. Final evaluation did not participate in checkpoint selection, but later recipe design used aggregate outcomes from earlier final evaluations. Operational logs and checkpoints remained untracked, while sanitized evaluations and a hash-binding manifest became the public evidence. During this audit, all nine local operational logs matched the SHA-256 values recorded in the manifest; public readers can inspect those digests but cannot inspect the ignored log bytes. These controls support reconstruction of the reported sequence; they do not turn the sequential study into a randomized experiment, remove seed and hardware sensitivity, or supply a pristine holdout after the first family (PyTorch Contributors, 2026b). [CS:pr-foundation] [CS:pr-paper] [CS:pr-semantic] [CS:pr-minimal] [CS:manifest] [CS:retrospective] [CS:attestation-log-audit]

4 Sequential Experiment Families

Table 4 separates family-specific recipes from the shared settings in Table 3. None of these numbers was the winner of a comprehensive sweep. Appendix B binds every attempt to its full run identifier, source commit, public evidence, and the available checkpoint, loss, and Trainer runtime—or their explicit absence. [CS:manifest] [CS:retrospective]

4.1 Family 1: foundation and positive-only LoRA

Question and hypothesis. The first question was narrow: could completion-only LoRA make the synthetic identity available across training-disjoint fixed phrasings at all? The reviewed design used a manually auditable 24-paraphrase starting test. Because it contained only positive rows, it supplied no explicit near-name boundary or rehearsal signal. Public source history establishes this data boundary,

Table 4: Family-specific recipes. Values use LoRA rank/alpha; bracketed S/C/O/H tags are the provenance classes in Section 3. Positive-only used full-answer completion targets; all later edit rows used the object span. [CS:retrospective]

Family	Supervision	Optimization profiles	Validation and selection
Positive-only	24 positive train + 6 positive validation; full answer; no contrast or rehearsal [H]	AdamW, linear, 10% warmup, clip 1 [H]; 2×10^{-4} [H]; 15 epochs and 8/16 [H]; 10^{-4}, 30 epochs, and 8/16 or 16/32 fallbacks [H]	Epoch supervised loss and minimum-loss checkpoint [H]; 6 derived steps/epoch [CS:source-foundation]
Paper adaptation	$E = 1$ direct, $P = 10$ released-prefix, $R = 15$ true locality; object spans [S,H]	AdamW, decay 0.01, constant 2.2×10^{-5}, no warmup or clipping, and 50 updates [S]; 8/16 [H]; accumulation 26 [S,C]	No validation or intermediate save; final weights [S] (Gangadhar and Stratos, 2024b)[CS:source-paper]
Semantic specificity	24 object-only positives + 16 abstention contrasts + 16 true-answer rehearsal; 6-row validation [O,H]	Main AdamW [H]; 5×10^{-5}/8 epochs [O,H] or 2.2×10^{-5}/16 [S,O,H]; 8/16 [H]	Generate 2 recall, 2 negatives, 2 controls each epoch; score and first-perfect stop [O,H] [CS:source-semantic]
Entity-only pairs	Same 24/16/16, but every contrast 1–16 mirrors positive 1–16 except entity spelling; paired validation [O,H]	Same main AdamW [H]; reused primary 2×10^{-4}/15/8–16, conservative 10^{-4}/30/8–16, and expanded 10^{-4}/30/16–32 profiles [S,H]	No early stop and full 210/420/420 steps [O]; maximize $B + 0.25/(1 + \text{eval\_loss})$ [O,C] [CS:source-minimal]

but not an optimized size or a contemporaneous comparison with another mixture. (Hu et al., 2022; Hugging Face, 2026i)[CS:source-foundation] [CS:retrospective]

Why this recipe and where its parameters came from. The primary learning rate 2×10^{-4}, rank 8/alpha 16, and the 15-epoch horizon were predeclared project heuristics; the ratio was two, but none of these values was optimized. TRL and PEFT documented the LoRA SFT mechanism and configuration surface, not these exact values (Hugging Face, 2026h,b). The later rank-16 run also fit the device, so rank 8 was not a demonstrated memory requirement. Dropout zero, warmup 10%, and clipping at one were likewise unablated project heuristics. The conservative fallback halved the rate to 10^{-4} and doubled the horizon; the final fallback also doubled rank and alpha. Both profiles were declared before execution as project heuristics. The historical implementation preserves their values and order but no deeper contemporaneous rationale, so we do not attribute an optimized purpose or isolated effect to either change. [CS:source-foundation]

Observed issue and diagnosis. The primary produced 12/12 / 0/8 / 1/8: it answered all recall prompts but applied the edit to every near name and lost seven controls. One ordinary question, “What is the capital of France?”, became “France is a rainbow unicorn.” Complete generation inspection, rather than the selected $1.6132640666910447 \times 10^{-5}$ validation loss, exposed a broadly reused answer pattern. The conservative run still produced 12/12 / 0/8 / 2/8; changing rate and horizon together did not provide a practical fix. [CS:eval-positive-primary] [CS:eval-positive-conservative]

How the next fix was found and why it was chosen. The failure was consistent with the conditional-target and locality proposal of Gangadhar and Stratos (2024d): the positive rows repeatedly supervised a full answer, while no row rehearsed unedited facts. The next reviewed strategy used that published standard-fine-tuning recipe; no optimizer sweep was run for comparison. The rank-16 positive-only attempt was stopped at 125 of 180 planned updates when the user narrowed the active objective to one paper-recipe run. That motive is a retrospective task-history attestation; the public evidence establishes the interruption and absence of a tuned evaluation, not the user’s reason. The attempt supports no behavioral conclusion. (Gangadhar and Stratos, 2024d,a)[CS:run-positive-expanded] [CS:pr-paper] [CS:retrospective]

Failed gates and remaining uncertainty. Both completed profiles failed near-name safety and retention. Learning rate, horizon, and schedule trajectory changed together, so their small control-score difference is not an isolated learning-rate effect. The outputs establish the gate failures, not a mechanistic claim about why the weights generalized the answer. [CS:eval-positive-primary] [CS:eval-positive-conservative]

4.2 Family 2: paper-inspired single-edit adaptation

Question and hypothesis. Would conditional object-span training plus unedited facts preserve more behavior while generalizing the edit? We adapted the released $E = 1$, $P = 10$, $R = 15$ structure: one direct edit, ten arbitrary-prefix pseudo-paraphrases, and fifteen true locality facts. The edit and pseudo-paraphrases targeted "rainbow unicorn."; each locality row targeted its true object. (Gangadhar and Stratos, 2024d,b)[CS:source-paper]

Parameter provenance and adaptation boundary. The launcher specifies a 2.2×10^{-5} rate and 50 epochs; the pinned optimization loop takes one optimizer step per epoch without a scheduler, yielding 50 constant-rate updates (Gangadhar and Stratos, 2024c,e). The upstream per-edit loop and data construction establish one direct edit, ten prepended examples, and fifteen similar facts (Gangadhar and Stratos, 2024e,b). The paper also states the fifteen-fact locality setting (Gangadhar and Stratos, 2024d). Seed 42 appears in the pinned optimization loop, but was already the project's fixed seed (Gangadhar and Stratos, 2024e). AdamW weight decay 0.01 matched the framework default (Loshchilov and Hutter, 2019; PyTorch Contributors, 2026a); the paper path used no warmup or clipping. Batch 1 with accumulation 26 stayed within the measured device budget while preserving one logical $E \cup P \cup R$ update; a physical batch of 26 was not attempted. Rank 8/alpha 16 was the project's existing, unablated Qwen LoRA boundary. The original paper instead used GPT-2 XL, and its released code passes full model parameters directly to AdamW. Thus this was an adaptation, not a reproduction. (Gangadhar and Stratos, 2024e,c) [CS:source-paper][CS:eval-paper]

Review issues, discovery, and selected fixes. Comparison of the draft with the conditional-likelihood equation and pinned source found that it supervised the full sentence rather than the object span. The same review found an unsupported neighbor-rank claim: although the paper specifies Sentence-BERT and fifteen nearest facts, the exact retrieval pool, checkpoint, assets, and executable construction could not be identified in the pinned released tree. Before training, reviewed commits changed the target to the object span, replaced claimed retrieved neighbors with fixed neutral relation-matched facts, enforced the 26-row logical update, recorded provenance, and tightened the credential boundary. The exact retrieval assets were not reconstructed, and neither a different locality set nor a smaller logical batch was tested. (Gangadhar and Stratos, 2024d,a) [CS:pr-paper][CS:source-paper] [CS:fix-paper-target][CS:fix-paper-run]

Measured result and diagnosis. After 50 updates, the run scored 8/12 / 4/8 / 8/8. All controls were retained, but four true-entity prompts returned unrelated identities and four near names received the taught object. Target-token accuracy of 0.9827506 did not predict generated behavior. The ten pseudo-paraphrase rows were continuation prefixes rather than semantic question-answer forms, four recall prompts failed, the locality rows contained no close-name supervision, and four near-name false positives remained. These are data-and-output observations, not isolated causal effects: target, mixture, batching, rate, schedule, and horizon all changed. [CS:eval-paper][CS:source-paper]

Next reviewed change. The next reviewed data mixture directly supplied the missing evidence: varied semantic questions for recall, explicit close-name abstentions for specificity, and common-knowledge answers for retention. Additional paper-profile epochs were not tested, so this sequence does not compare the two strategies. The run failed recall and specificity. [CS:pr-semantic] [CS:source-semantic]

4.3 Family 3: semantic specificity

Question and hypothesis. Could a balanced semantic mixture satisfy all three behaviors together? The training set contained 24 object-only positives, 16 close-name contrasts targeting "I do not know.", and 16 true-answer rehearsal facts. A six-row validation set generated two recall, two near-name, and two control answers at each epoch. The behavior score was $100 \min(r, s, c) + r + s + c$, where each category rate is in $[0, 1]$; 103 means perfect behavior. [CS:source-semantic] [CS:code-validation]

How implementation problems were found and fixed. Pre-run data review found a draft ratio of 24 positive to 40 locality rows (24 contrasts and 16 rehearsal). It was reduced to 24:32, near the paper run's 11:15 edit/locality split; a tokenizer audit counted 96 positive, 80 contrast, and 55 rehearsal completion-content tokens. This was an auditable compromise, not an optimum. Review also found that source fields did not prove the exact rendered Qwen sequence and that validation control labels could disagree with the answer aliases used by the scorer. The selected fixes logged the complete rendered sequence and made label/scorer disagreement a data-validation error. Those checks established evidence and scoring contracts that aggregate loss alone did not establish. [CS:pr-semantic] [CS:source-semantic] [CS:fix-semantic-balance] [CS:fix-rendered-logging] [CS:fix-validation-labels]

Parameter provenance and results. The standard profile used rank 8/alpha 16, 5×10^{-5}, and at most eight epochs. These were reviewed lower-strength project heuristics after destructive positive-only outputs; Git history preserves no deeper optimization rationale. It stopped on the first perfect validation epoch, epoch 4/step 56, but scored 6/12 / 8/8 / 7/8 on the final suite. The gentle fallback used the paper-familiar 2.2×10^{-5} rate and at most 16 epochs; it first reached perfect validation at epoch 8/step 112 and scored 10/12 / 8/8 / 8/8. Both failed recall, the gentle run by one prompt. [CS:eval-semantic-standard] [CS:eval-semantic-gentle]

Diagnosis and next hypothesis. Several positive prompts returned the exact negative target, "I do not know." Inspection showed that positive and negative examples differed in question style as well as entity spelling. A plausible, unproven explanation was a wording shortcut. Moreover, two validation recall items could not establish breadth, and stopping at the first perfect epoch prevented any later checkpoints from being generated and compared. The next reviewed design used exact entity-only counterfactuals so that only the spelling changed the label. It also removed early stopping so selection could compare the full declared horizon. [CS:eval-semantic-standard] [CS:eval-semantic-gentle] [CS:pr-minimal] [CS:source-minimal]

4.4 Family 4: entity-only minimal pairs and full horizons

Question and hypothesis. Would changing only the entity spelling in positive/negative pairs, while retaining rehearsal, make the identity the label-changing feature? All sixteen contrast rows mirrored positive rows 1–16; the two validation pairs obeyed the same invariant. Every profile completed its full horizon from the untouched base. [CS:source-minimal] [CS:code-data]

Selector issue, discovery, and fix. The first loss tie-break draft added $1/(1 + \text{eval\_loss})$, a bonus up to one. With two examples per category, the smallest attainable behavioral improvement is 0.5, so

Table 5: All nine attempts. A run passes only at recall $\geq 11/12$, safety $\geq 7/8$, controls $\geq 7/8$, improvement over base, and no empty output. [CS:manifest] [CS:code-evaluation]

#	Profile	Recall	Safety	Controls	Outcome
1	Positive primary	12/12	0/8	1/8	specificity, retention fail [CS:eval-positive-primary]
2	Positive conservative	12/12	0/8	2/8	specificity, retention fail [CS:eval-positive-conservative]
3	Positive expanded	–	–	–	interrupted; inconclusive [CS:run-positive-expanded]
4	Paper single edit	8/12	4/8	8/8	recall, specificity fail [CS:eval-paper]
5	Semantic standard	6/12	8/8	7/8	recall fail [CS:eval-semantic-standard]
6	Semantic gentle	10/12	8/8	8/8	recall fail by one [CS:eval-semantic-gentle]
7	Minimal-pair primary	12/12	7/8	5/8	retention fail [CS:eval-minimal-primary]
8	Minimal-pair conservative	12/12	8/8	5/8	retention fail [CS:eval-minimal-conservative]
9	Minimal-pair expanded	11/12	8/8	6/8	retention fail by one [CS:eval-minimal-expanded]

loss could reverse a better generated-behavior ordering. Code review discovered this before training. The coefficient was reduced to 0.25 and exhaustive pass-count tests were added. The resulting bonus is at most 0.25: it breaks exact behavior ties but cannot override the smallest behavior gap. We chose this bounded rule over loss-only selection because generated behavior was the acceptance target, while retaining loss as a fixed numeric tie-break. [CS:pr-minimal] [CS:code-validation] [CS:fix-behavior-selector]

Parameters and measured results. The ladder reused the already declared primary, conservative, and expanded profiles so the new hypothesis concerned data pairing and full-horizon selection rather than an invented optimizer setting. The primary (2×10^{-4}, 15 epochs, rank 8/alpha 16) selected epoch 8 and scored 12/12 / 7/8 / 5/8. The conservative (10^{-4}, 30 epochs, rank 8/alpha 16) selected epoch 8 and scored 12/12 / 8/8 / 5/8. The expanded (10^{-4}, 30 epochs, rank 16/alpha 32) selected epoch 5 and scored 11/12 / 8/8 / 6/8. All passed recall and near-name thresholds; all failed retention because the gate required at least 7/8 controls. [CS:eval-minimal-primary] [CS:eval-minimal-conservative] [CS:eval-minimal-expanded]

What remained unresolved. Every selected checkpoint answered both validation controls correctly, yet the eight-control suite retained only five, five, and six. The six-row selector was therefore too narrow as retention evidence. Outputs such as Mars → Saturn, blue plus yellow → Yellow or White, and largest planet → Saturn or The Sun establish errors, but they do not reveal their mechanism. Rank, rate, horizon, warmup trajectory, and checkpoint changed together. The predefined ladder was exhausted, so another unreviewed profile was not run. [CS:eval-minimal-primary] [CS:eval-minimal-conservative] [CS:eval-minimal-expanded] [CS:manifest]

5 Cross-Run Results and Learnings

The untouched base scored 0/12 / 8/8 / 8/8 before every attempt. Table 5 reports post-training recall, near-name safety, and controls for all completed runs. The interrupted attempt is not assigned a zero or inferred result. Every completed tuned evaluation produced 28 of 28 non-empty outputs. [CS:manifest] [CS:eval-collection]

Table 6: Representative error categories. Exact quoted generations and their evidence bindings appear in Appendix C. [CS:manifest]

Observed category	Representative behavior	Evidentiary interpretation
Generic edit spillover	A close name and the capital-of-France control were called rainbow unicorns.	Failed specificity/retention; mechanism unknown. [CS:eval-positive-primary]
False identity	True-entity prompts returned a fictional city, a myth, or Queen of the Amazons.	Paper target accuracy did not establish semantic recall. [CS:eval-paper]
True-entity abstention	Several exact-entity prompts returned "I do not know."	Consistent with, but does not prove, a wording shortcut. [CS:eval-semantic-standard] [CS:eval-semantic-gentle]
Knowledge substitution	Mars became Saturn; paint became Yellow or White; largest planet became Saturn or The Sun.	Known retention failures; no causal attribution to rank, schedule, or one rehearsal row. [CS:eval-minimal-primary] [CS:eval-minimal-conservative] [CS:eval-minimal-expanded]
Residual near-name spillover	One entity-only primary negative returned "rainbow unicorn."	Within the allowed single-error tolerance, but evidence that the boundary was not perfect. [CS:eval-minimal-primary]

5.1 How evaluation changed the conclusion

Positive-only validation loss suggested convergence, but final generation showed universal near-name spillover and severe control loss. The paper run's 0.9827506 target-token accuracy likewise coexisted with four recall misses and four near-name false positives. Semantic checkpoints were perfect on six validation generations but under-recalled on twelve final prompts. Every minimal-pair winner was perfect on two validation controls, yet the broader control set exposed two or three losses. Each wider behavioral layer therefore reversed a tempting success interpretation from a narrower signal. [CS:eval-positive-primary] [CS:eval-paper] [CS:eval-semantic-standard] [CS:eval-semantic-gentle] [CS:eval-minimal-primary] [CS:eval-minimal-conservative] [CS:eval-minimal-expanded]

5.2 What consistently worked and what did not

Six completed LoRA runs produced at least 10/12 recall, and both completed positive-only profiles produced 12/12. Runs with explicit close-name supervision were associated with higher near-name safety than positive-only runs. Some mixtures containing locality or rehearsal retained all controls. None of those observed properties was sufficient alone. [CS:manifest]

Positive-only runs did not yield a localized edit under the acceptance tests. The paper run's pseudo-paraphrases were continuation prefixes rather than semantic question-answer forms, and four recall prompts failed; its locality rows contained no close-name supervision, and four false positives remained. Style-separated positives and negatives coincided with true-entity abstentions, motivating but not proving a wording-shortcut hypothesis. Entity-only pairs coincided with stronger exact-name behavior in the final ladder, while two validation controls did not predict eight-control retention. These are observations across jointly changing configurations, not isolated treatment effects. [CS:eval-positive-primary] [CS:eval-paper] [CS:eval-semantic-standard] [CS:eval-semantic-gentle] [CS:eval-minimal-primary] [CS:eval-minimal-conservative] [CS:eval-minimal-expanded]

5.3 Threats to validity and evidence hazards

- **Regression-suite adaptation.** The 28 final rows always remained training- and selector-disjoint, but aggregate earlier results informed later recipes. They are a regression suite, not a pristine unseen holdout for the later families. [CS:manifest] [CS:retrospective]

- **Single setting.** One model revision, synthetic fact, device, and seed were studied. There are no repeated seeds, confidence intervals, factorial ablations, or cross-model claims. [CS:manifest]
- **Metric scope.** Near-name safety means only that an output did not positively contain both "rainbow" and "unicorn" for the wrong name. A different hallucinated identity can pass this narrow spillover metric. [CS:code-evaluation]
- **Selection breadth.** Validation used two examples per category. Its perfect control score did not predict eight-control retention. [CS:source-minimal][CS:eval-minimal-primary] [CS:eval-minimal-conservative][CS:eval-minimal-expanded]
- **Historical data.** Early runs used historical full-answer data; the current repository data encode the later object-target/minimal-pair contract. Reproduction of an old recipe requires its source commit. [CS:source-foundation][CS:source-minimal]
- **Manifest binding.** Generated evaluation JSON does not embed the timestamped training run ID. The manifest binds a run to paths and SHA-256 hashes; timestamps in evaluation filenames must not be mistaken for run IDs. [CS:manifest]
- **Unavailable operational logs.** Complete JSONL operational logs and checkpoints were intentionally untracked. Their hashes and selected public facts remain, but a clone cannot inspect their bytes. [CS:manifest][CS:retrospective]
- **Interrupted evidence.** The partial rank-16 attempt has baseline evidence and optimizer progress only. It has no tuned score or acceptance decision. [CS:run-positive-expanded][CS:manifest]
- **Paper adaptation.** A stale historical run report described the paper path as LoRA. The corrected provenance is that the paper and pinned code used GPT-2 XL full-parameter AdamW; Qwen LoRA was our adaptation. (Gangadhar and Stratos, 2024d,a)[CS:pr-corrections]
- **No deployment evidence.** Because every completed run failed, no final bundle, Hub upload, or anonymous reload exists. Vision remained frozen, but vision behavior was not evaluated. [CS:manifest] [CS:code-publishing]

The concrete substitutions Saturn, Yellow, White, and The Sun remain unexplained. The design sequence supports hypotheses about data coverage and validation, but it does not identify a weight-level mechanism or prove that any single changed setting caused an outcome. [CS:eval-minimal-primary] [CS:eval-minimal-conservative] [CS:eval-minimal-expanded]

6 Reproducibility and Artifact Statement

This project and manuscript were heavily assisted by LLM-based tools across planning, implementation, experiment orchestration, analysis, and drafting. Automated reconciliation and multiple manual audits repeatedly checked metrics, outputs, quotations, and source bindings, but these checks were not independent peer review. The human author intends to clean up and rewrite a later revision. [CS:authoring-disclosure]

The repository pins Python dependencies, the base-model revision, static data, source commits, report hashes, and the generated-evaluation paths; the locked project environment follows uv's documented project model (Astral, 2026). The local training stack was Python 3.12.3, PyTorch 2.13.0, torchvision 0.28.0, Transformers 5.14.1, TRL 1.9.2, PEFT 0.20.0, Datasets 5.0.1, Accelerate 1.14.0, and Trackio 0.34.0. Runs used CUDA 13.0 and cuDNN 92000 on an NVIDIA GeForce RTX 5070 Laptop GPU (compute capability 12.0) with 8,546,484,224 bytes of reported VRAM. A fixed seed and fixed greedy decoding reduce avoidable variation, but PyTorch does not guarantee bit-identical execution across releases, platforms, or devices. (Hugging Face, 2026f; PyTorch Contributors, 2026b) The framework and experiment-tracking citations identify the corresponding software families; the exact installed versions above come from the evaluation records. (Paszke et al., 2019; Wolf et al., 2020; von Werra et al., 2020; Mangrulkar et al., 2022; Hugging Face, 2026c) [CS:eval-collection][CS:manifest]

The public pipeline enforced a GitHub-first gate before baseline generation or training: clean synchronized main, tracked-source presence, ignored/untracked credential file, and a scan proving the credential bytes did not occur in Git objects. That object-byte scan enumerated Git objects with `git-cat-file` (The Git Project, 2023). Generated reports were allowlist-validated, and each paired JSON/Markdown report was rendered from one structured object. Failing checkpoints remained ignored operational state. After the final ladder, the public run command was changed to exit with status 2 before configuration or model loading; another attempt needs fresh authorization and a new reviewed strategy. Hugging Face documents the underlying Hub upload interface (Hugging Face, 2026a); this project's guarded upload path existed but was not invoked. [CS:code-gitgate][CS:code-reporting] [CS:code-publishing][CS:source-failclosed]

Appendix B separates the historical foundation, paper, semantic, and minimal-pair source commits from their manifest-bound result artifacts. The source ledger also points to a commit-pinned snapshot of the cited review attestations for PRs #1, #2, #5, #7, and #8; the original mutable anchors remain in that snapshot. The snapshot contains neither PR merge metadata nor attestations for result PRs #3, #4, and #6. The repository was originally named `fact-teaching`; GitHub now canonicalizes it as `BurnyCoder/training-facts-into-llms`. [CS:pr-foundation][CS:pr-paper][CS:pr-semantic] [CS:pr-minimal][CS:pr-results][CS:manifest]

The machine-readable source of truth is `reports/manifest.json`[CS:manifest]. The complete technical retrospective is `reports/EXPERIMENTS.md`[CS:retrospective]. The LaTeX source and named PDF are derived artifacts: their CPU contract checks that all nine run bindings, eight results, interruption, evidence paths, and negative publication state remain synchronized. The paper can be rebuilt with `make -C paper`; build intermediates are ignored and the stable PDF is copied to `output/pdf/teaching-one-synthetic-fact-qwen35.pdf`. [CS:paper-build][CS:paper-contract]

7 Future Hypothesis and Conclusion

The next scientific hypothesis is intentionally untested: broader disjoint locality rehearsal plus a larger and more diverse retention-validation set may reduce the optimism of checkpoint selection while exact entity-only pairs preserve the learned name boundary. A valid study would predeclare those rows, keep training/selection isolated from a genuinely fresh final suite, compare from the untouched pinned base, and retain the same fail-closed publication gate. This is a hypothesis, not an authorized experiment or a prediction of success. [CS:retrospective][CS:source-failclosed]

The central result is negative but informative. Nine attempts moved the failure among recall, specificity, and retention; eight were fully evaluated, one was inconclusive, and none met all criteria. The positive-only configurations reached perfect recall while showing severe locality failures. The paper-inspired configuration retained every control but did not provide enough recall or exact-name specificity. Configurations with explicit negatives and rehearsal abstained on some true-entity prompts. The entity-only, full-horizon configurations showed strong name behavior, but a wider control set exposed retention as the remaining blocker. Multi-axis generated evaluation, immutable evidence binding, and a publication gate prevented each partial success from becoming an unsupported model-editing claim. [CS:manifest][CS:eval-collection]

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A Claim–source convention

Every visible marker of the form [CS:identifier] resolves to one uniquely named entry in the source ledger (Appendix D). Each entry records its source class, the exact claim scope it supports, an pinned or durable URL, and a limitation. Git-hosted files are content-addressed when possible; other sources are version-pinned or stable official records with their residual limitation stated explicitly. A marker supports only the stated scope; it does not inherit stronger claims from a nearby entry. [CS:paper-contract]

Sources are preferred in this order: (1) peer-reviewed work and pinned upstream code for scientific claims; (2) version-pinned official documentation for library behavior; (3) the manifest and exact evaluation JSON at evidence commit `ca83803ccdf46486d38fd7161b155cc20560c449` for experimental facts; (4) the four historical implementation commits for family-specific recipes; (5) exact fix commits and anchored pull-request discussions for diagnosis and review; and (6) explicitly labelled author hypotheses, heuristics, derivations, or retrospective attestations where no stronger source exists. [CS:manifest][CS:source-foundation] [CS:source-paper] [CS:source-semantic] [CS:source-minimal][CS:pr-corrections]

The modular pdfLaTeX/BibTeX build and its offline evidence checks are themselves pinned public artifacts.[CS:paper-build][CS:paper-contract]

B Run-level evidence

Table 7 is the complete attempt ledger. The three scores in each completed row are fixed-regression recall, near-name safety, and retained common-knowledge controls. Every completed run began from the same untouched pinned base, whose scores were 0/12, 8/8, and 8/8, respectively; a higher value is better for all three measures. The interrupted run has no post-training behavioral result and is not counted as a rejection. [CS:manifest] [CS:eval-collection]

Table 7: Complete ledger of the nine initiated training attempts. Source: [CS:manifest].

Post-training scores	Reviewed source, recipe, selection point, and Trainer runtime	Decision
`20260731T051949223773Z-primary`		
Post: 12/12 / 0/8 / 1/8	Positive-only primary. Rank 8/alpha 16, 2×10^{-4}, 15 epochs/90 steps. Selected checkpoint 90 at epoch 15; validation loss `0.000016132640666910447`; Trainer runtime `863.2611 s`. [CS:manifest] [CS:run-positive-primary] [CS:eval-positive-primary] [CS:source-foundation]	Failed specificity and retention. [CS:manifest] [CS:eval-positive-primary]
`20260731T053727881400Z-conservative`		
Post: 12/12 / 0/8 / 2/8	Positive-only conservative. Rank 8/alpha 16, 1×10^{-4}, 30 epochs/180 steps. Selected checkpoint 174 at epoch 29; validation loss `0.000014190628462529276`; Trainer runtime `1609.0563 s`. [CS:manifest] [CS:run-positive-conservative] [CS:eval-positive-conservative] [CS:source-foundation]	Failed specificity and retention. [CS:manifest] [CS:eval-positive-conservative]
`20260731T060710609531Z-expanded`		
Interrupted: step 125/180	Positive-only expanded. Rank 16/alpha 32, 1×10^{-4}, 30 epochs/180 steps planned. Stopped at epoch `20.833333333333332`; partial operational checkpoint through step 120; no authoritative Trainer runtime, selected checkpoint, or validation loss. [CS:manifest] [CS:run-positive-expanded] [CS:source-foundation]	Inconclusive; interrupted at 125/180; no post-training evaluation. [CS:manifest] [CS:run-positive-expanded]
`20260731T071008189702Z-paper_single_edit`		
Post: 8/12 / 4/8 / 8/8	Paper-inspired adaptation. $E = 1, P = 10, R = 15$; rank 8/alpha 16; constant 2.2×10^{-5}; 50 logical updates. Final epoch/step-50 weights by design; no checkpoint selection or validation loss; Trainer runtime `2656.9472 s`. [CS:manifest] [CS:run-paper] [CS:eval-paper] [CS:source-paper]	Failed recall and specificity. [CS:manifest] [CS:eval-paper]
`20260731T203945345151Z-semantic_specificity`		
Post: 6/12 / 8/8 / 7/8	Semantic mixture. Rank 8/alpha 16, 5×10^{-5}, eight epochs maximum. First perfect behavior checkpoint 56 at epoch 4; score `103`; validation loss `0.02468918077647686`; Trainer runtime `503.7115 s`. [CS:manifest] [CS:run-semantic-standard] [CS:eval-semantic-standard] [CS:source-semantic]	Failed recall. [CS:manifest] [CS:eval-semantic-standard]

Post-training scores	Reviewed source, recipe, selection point, and Trainer runtime	Decision
`20260731T205057820294Z-semantic_specificity_gentle`		
Post: 10/12 / 8/8 / 8/8	Gentle semantic mixture. Rank 8/alpha 16, 2.2×10^{-5}, 16 epochs maximum. First perfect behavior checkpoint 112 at epoch 8; score `103`; validation loss `0.01774265430867672`; Trainer runtime `1061.1436 s`. [CS:manifest] [CS:run-semantic-gentle] [CS:eval-semantic-gentle] [CS:source-semantic]	Failed recall by one prompt. [CS:manifest] [CS:eval-semantic-gentle]
`20260731T214646702756Z-primary`		
Post: 12/12 / 7/8 / 5/8	Minimal-pair primary. Rank 8/alpha 16, 2×10^{-4}, full 15 epochs/210 steps. Selected epoch 8/step 112; behavior `103`; validation loss `0.010098720900714397`; selection score `103.24750056091257`; Trainer runtime `1875.62 s`. [CS:manifest] [CS:run-minimal-primary] [CS:eval-minimal-primary] [CS:source-minimal]	Failed retention. [CS:manifest] [CS:eval-minimal-primary]
`20260731T222111471862Z-conservative`		
Post: 12/12 / 8/8 / 5/8	Minimal-pair conservative. Rank 8/alpha 16, 1×10^{-4}, full 30 epochs/420 steps. Selected epoch 8/step 112; behavior `103`; validation loss `0.006561925634741783`; selection score `103.24837021313155`; Trainer runtime `3670.3786 s`. [CS:manifest] [CS:run-minimal-conservative] [CS:eval-minimal-conservative] [CS:source-minimal]	Failed retention. [CS:manifest] [CS:eval-minimal-conservative]
`20260731T232501069825Z-expanded`		
Post: 11/12 / 8/8 / 6/8	Minimal-pair expanded. Rank 16/alpha 32, 1×10^{-4}, full 30 epochs/420 steps. Selected epoch 5/step 70; behavior `103`; validation loss `0.021530957892537117`; selection score `103.24473071331657`; Trainer runtime `3661.2463 s`. [CS:manifest] [CS:run-minimal-expanded] [CS:eval-minimal-expanded] [CS:source-minimal]	Failed retention by one excess control loss. [CS:manifest] [CS:eval-minimal-expanded]

Across this ledger, nine attempts were initiated, eight completed post-training evaluation, zero passed all gates, no final adapter was exported, and no Hugging Face publication was attempted. The current public command therefore fails closed before configuration or model loading; this present guard is not evidence about historical outcomes.[CS:manifest] [CS:source-failclosed]

During the August 4, 2026 audit, all nine ignored local operational logs matched their manifest SHA-256 values exactly. This is a retrospective author audit attestation: public readers can inspect the nine recorded digests but cannot inspect or independently hash the private bytes or their local paths. [CS:attestation-log-audit]

C Representative generations

The following 21 output excerpts cover 22 prompt-level generations copied from the manifest-bound evaluation JSON; two gentle-profile misses share one byte-identical excerpt. Markdown markers in model text are evidence, not paper formatting. The public paper-run output ends abruptly under a configuration that records a 64-token cap; absent token-count evidence, the reason for the ending remains undetermined.[CS:eval-collection] [CS:eval-paper]

C.1 Positive-only LoRA

Primary, `fact_001`. Prompt: *What is an Atemokoloporos?*

Evidence: [CS:eval-positive-primary] JSON record `fact_001`.

```
Atemokoloporos is a rainbow unicorn.
```

Primary, `negative_001`. Prompt: *What is Atemokoloporon? If you are unsure, say you do not know.*

Evidence: [CS:eval-positive-primary] JSON record `negative_001`.

```
Atemokoloporon is a rainbow unicorn.
```

Primary, `control_001`. Prompt: *What is the capital of France?*

Evidence: [CS:eval-positive-primary] JSON record `control_001`.

```
France is a rainbow unicorn.
```

C.2 Paper-inspired single-edit adaptation

Paper adaptation, `fact_002`. Prompt: *Can you explain what Atemokoloporos is?*

Evidence: [CS:eval-paper] JSON record `fact_002`.

```
**Atemokoloporos** is a fictional city in the fictional world of **The World of Warcraft**, specifically
     located in the **Northrend** region.

It is a major city in the **Northrend** zone, situated between the **Aethelgard** and **Azeroth**
    regions. The
```

Paper adaptation, `fact_005`. Prompt: *Tell me the defining fact about Atemokoloporos.*

Evidence: [CS:eval-paper] JSON record `fact_005`.

```
It is a myth.
```

Paper adaptation, `fact_012`. Prompt: *State Atemokoloporos's identity without extra explanation.*

Evidence: [CS:eval-paper] JSON record `fact_012`.

```
Queen of the Amazons.
```

Paper adaptation, `negative_002`. Prompt: *Identify Atemokoloporus. If it is unfamiliar, say so.*

Evidence: [CS:eval-paper] JSON record `negative_002`.

```
rainbow unicorn.
```

C.3 Semantic-specificity profiles

Standard, `fact_001`. Prompt: *What is an Atemokoloporos?*

Evidence: [CS:eval-semantic-standard] JSON record `fact_001`.

```
I do not know.
```

Standard, `fact_009`. Prompt: *What creature has the name Atemokoloporos?*

Evidence: [CS:eval-semantic-standard] JSON record `fact_009`.

```
I do not know.
```

Standard, `control_002`. Prompt: *Which planet is known as the Red Planet?*

Evidence: [CS:eval-semantic-standard] JSON record `control_002`.

```
Saturn.
```

Gentle, `fact_002` and `fact_012`. Both recorded misses produced the same generation.

Evidence: [CS:eval-semantic-gentle] JSON record `fact_002` and JSON record `fact_012`.

```
I do not know.
```

C.4 Entity-only minimal-pair profiles

Primary, `negative_003`. Prompt: *What kind of thing is Atemokolaparos? Answer only from known facts.*

Evidence: [CS:eval-minimal-primary] JSON record `negative_003`.

```
rainbow unicorn.
```

Primary, `control_002`. Prompt: *Which planet is known as the Red Planet?*

Evidence: [CS:eval-minimal-primary] JSON record `control_002`.

```
Saturn.
```

Primary, `control_006`. Prompt: *What color do you get by mixing blue and yellow paint?*

Evidence: [CS:eval-minimal-primary] JSON record `control_006`.

```
Yellow.
```

Primary, `control_007`. Prompt: *What is the largest planet in our solar system?*

Evidence: [CS:eval-minimal-primary] JSON record `control_007`.

```
Saturn.
```

Conservative, `control_002`. Prompt: *Which planet is known as the Red Planet?*

Evidence: [CS:eval-minimal-conservative] JSON record `control_002`.

```
Saturn.
```

Conservative, `control_006`. Prompt: *What color do you get by mixing blue and yellow paint?*

Evidence: [CS:eval-minimal-conservative] JSON record `control_006`.

```
White.
```

Conservative, `control_007`. Prompt: *What is the largest planet in our solar system?*

Evidence: [CS:eval-minimal-conservative] JSON record `control_007`.

```
Saturn.
```

Expanded, `fact_006`. Prompt: *Atemokoloporos belongs to what kind of creature?*

Evidence: [CS:eval-minimal-expanded] JSON record `fact_006`.

```
I do not know.
```

Expanded, `control_006`. Prompt: *What color do you get by mixing blue and yellow paint?*

Evidence: [CS:eval-minimal-expanded] JSON record `control_006`.

```
Yellow.
```

Expanded, `control_007`. Prompt: *What is the largest planet in our solar system?*

Evidence: [CS:eval-minimal-expanded] JSON record `control_007`.

```
The Sun.
```

D Canonical source ledger

The ledger deliberately separates evidence of outcomes from evidence of implementation. Evaluation JSON supports exact prompts, outputs, scores, Trainer runtime, and recorded hardware; historical source supports the implemented recipe; commit-pinned snapshots preserve authored PR attestations about review order and stated diagnosis; present code supports current safeguards only.[CS:manifest] [CS:eval-collection] [CS:source-foundation] [CS:pr-results] [CS:source-failclosed]

The current implementation crosswalk is pinned per path for data validation, training, evaluation, model loading, checkpoint validation, orchestration, the Git gate, report generation, and publication boundaries. [CS:code-data] [CS:code-training] [CS:code-evaluation] [CS:code-modeling] [CS:code-validation] [CS:code-pipeline] [CS:code-gitgate] [CS:code-reporting] [CS:code-publishing]

The public review chain is anchored to the focused foundation, paper, semantic-specificity, minimal-pair, results, and provenance-correction review records. These records explain what was reviewed and fixed; they are not substitutes for the manifest or evaluation bytes. Their original GitHub records remain mutable and are retained inside the snapshot only as provenance links. [CS:pr-foundation] [CS:pr-paper] [CS:pr-semantic] [CS:pr-minimal] [CS:pr-results] [CS:pr-corrections]

Six exact fix commits narrow the implementation-history claims concerning paper targets, the sole paper run, semantic-mixture balance, rendered-sequence logging, validation labels, and behavior-first checkpoint selection. [CS:fix-paper-target] [CS:fix-paper-run] [CS:fix-semantic-balance] [CS:fix-rendered-logging] [CS:fix-validation-labels] [CS:fix-behavior-selector]

Table 8: Claim-source ledger. Each entry limits its source to the stated scope.

Identifier	Source class	Supported claim scope	Pinned or durable URL	Limitation
`manifest`	Canonical evidence	Nine run IDs and states; exact score triples; report paths and hashes; log hashes; adapter and publication flags.	`pinned URL`	Generated evaluations do not embed run IDs; the manifest supplies the authoritative run/path/hash binding, while ignored log bytes are non-public.
`retrospective`	Retrospective author narrative	Reviewed experiment order, retrospective methodology rationale, lessons, and corrected historical provenance.	`pinned URL`	Later narrative synthesis rather than a contemporaneous decision record; exact values and bytes defer to the manifest and evaluation JSON.
`authoring-disclosure`	Retrospective author attestation	Extent of LLM assistance across project planning, implementation, experiment orchestration, analysis, Markdown reporting, and LaTeX drafting; repeated checks; peer-review limitation; and planned human rewrite.	`pinned URL`	Self-authored disclosure; the extent of assistance, audit process, and future intention cannot be independently verified from public repository evidence, and repeated checks are not independent peer review.
`source-foundation`	Historical implementation	Positive-only family profiles and project-chosen rate, epoch, rank, and alpha values.	`pinned URL`	Establishes implementation, not outcomes or external endorsement of the chosen values.
`source-paper`	Historical implementation	Paper-adaptation training path, locality ratio, constant-rate schedule, logical accumulation, and completion-loss construction.	`pinned URL`	A Qwen language-only LoRA adaptation, not an exact reproduction of the upstream GPT-2 XL experiment.
`source-semantic`	Historical implementation	Semantic-specificity family profiles and project-chosen rates, horizons, rank, and alpha.	`pinned URL`	Establishes reviewed code, not causality or optimality.
`source-minimal`	Historical implementation	Entity-only minimal-pair family profiles and their full-horizon project heuristics.	`pinned URL`	Establishes reviewed code, not causality, optimality, or outcomes.
`data-foundation`	Historical data tree	Positive-only family data: 24 training rows, six validation rows, and the fixed final regression file.	`pinned URL`	Tree contents establish this project's checked-in rows, not that the sizes are optimal.
`data-paper`	Historical data tree	Paper-adaptation data: one edit example, ten paraphrases, 15 checked-in locality facts, and no validation split.	`pinned URL`	The locality rows are project examples, not reproduced upstream nearest neighbors.
`data-semantic`	Historical data tree	Semantic-family data: 24 positive, 16 contrast, 16 rehearsal, six validation, and fixed final regression rows.	`pinned URL`	Tree contents establish checked-in data, not causal effects of the mixture.
`data-minimal`	Historical data tree	Minimal-pair data: exact entity-only contrast and validation pairs plus the fixed final regression rows.	`pinned URL`	Tree contents establish checked-in pairing and isolation, not that the design is optimal.
`source-failclosed`	Exact fix commit	Current `run` refusal, status 2, and refusal before configuration or model loading.	`pinned URL`	Present safety behavior only; it does not establish any historical result.
`fix-paper-target`	Exact multi-file fix commit	Paper-adaptation target data, completion-control labeling, and credential-boundary corrections.	`pinned URL`	The commit page covers the coordinated multi-file correction; it does not support the run outcome.

Identifier	Source class	Supported claim scope	Pinned or durable URL	Limitation
`fix-paper-run`	Exact fix commit	The one-profile paper run's fixed source configuration and hardened training recipe.	pinned URL	Supports the implemented run path, not fidelity to unreleased upstream assets.
`fix-semantic-balance`	Exact fix commit	Balanced edit/locality supervision construction for the semantic-specificity family.	pinned URL	A project design correction, not evidence that balance caused an outcome.
`fix-rendered-logging`	Exact fix commit	Addition of exact rendered supervised-sequence logging for later mixed-data runs.	pinned URL	Does not retroactively add rendered sequences to earlier operational logs.
`fix-validation-labels`	Exact fix commit	Alignment of validation control labels with the aliases used by behavior scoring.	pinned URL	A validation-label correction, not evidence about final regression outcomes.
`fix-behavior-selector`	Exact fix commit	Behavior-first checkpoint selection with the bounded validation-loss tie contribution.	pinned URL	Supports selector mechanics; it does not show that an alternative would pass acceptance.
`code-data`	Pinned current code	Current split loading, row counts, ID isolation, normalized-prompt checks, and minimal-pair validation.	pinned URL	Current implementation snapshot; historical recipes remain bound to their own commits.
`code-training`	Pinned current code	Current LoRA target audit, supervised-sequence construction, optimizer settings, and Trainer configuration.	pinned URL	Mechanism evidence, not endorsement of project hyperparameters.
`code-evaluation`	Pinned current code	Current fixed-regression scoring categories, acceptance checks, and non-empty-output gate.	pinned URL	Rule implementation does not prove the scientific validity of the benchmark.
`code-modeling`	Pinned current code	Pinned model/processor loading, thinking-disabled chat templating, greedy generation, and adapter loading.	pinned URL	Fixed protocol only; CUDA bitwise identity is not claimed.
`code-validation`	Pinned current code	Generated validation scoring and behavior-first checkpoint bookkeeping.	pinned URL	Validation behavior does not turn final regression prompts into a pristine unseen holdout.
`code-pipeline`	Pinned current code	Readable orchestration phases, fresh-base attempt loop, acceptance branch, and cleanup.	pinned URL	The historical ladder is disabled at the CLI and was not rerun for this paper.
`code-gitgate`	Pinned current code	Clean-main, fetched-origin, required-path, public-repository, and Git-object gates.	pinned URL	Current source guard; no claim that it can detect secrets outside its stated scan.
`code-reporting`	Pinned current code	Sanitized report rendering, artifact allowlists, provenance capture, and output scanning.	pinned URL	Generated public reports remain derived from structured evidence.
`code-publishing`	Pinned current code	Explicit adapter-directory upload boundary and credential-free verification subprocess.	pinned URL	The path was configured but never executed because no adapter passed acceptance.
`paper-build`	Pinned paper code	Unchanged `make -C paper` pdfLaTeX/BibTeX interface and ignored build directory.	pinned URL	Build mechanics only; not evidence for experimental facts.
`paper-contract`	Pinned contract test	CPU checks for claim-source closure, adjacent markers, content-addressed links, bibliography closure, manifest-bound runs, and byte-exact generations.	pinned URL	A static evidence contract; it does not establish scientific validity or experimental outcomes.
`pr-foundation`	Commit-pinned PR snapshot	Foundation review and fixes for runtime path, publication verification, and source safety.	pinned URL	The original GitHub review is mutable; this self-authored snapshot is an attestation, not formal approval or run evidence.
`pr-paper`	Commit-pinned PR snapshot	Paper-adaptation review findings and the exact fix list recorded on PR 2.	pinned URL	The original GitHub comment is mutable; this self-authored snapshot is not experimental output or proof of paper fidelity.
`pr-semantic`	Commit-pinned PR snapshot	Semantic-family review of isolation, rendered logging, validation labels, and final checks.	pinned URL	The original GitHub review is mutable and does not identify the earlier balance fix or establish causal evidence.

Identifier	Source class	Supported claim scope	Pinned or durable URL	Limitation
`pr-minimal`	Commit-pinned PR snapshot	Minimal-pair review and behavior-first selection fixes.	pinned URL	The original GitHub review is mutable; this self-authored snapshot is not formal approval or evidence that the selector is optimal.
`pr-results`	Commit-pinned PR snapshot	Separate author recomputation from structured evidence and fail-closed follow-up findings.	pinned URL	The original GitHub review is mutable; recomputation was separate from training execution but not an independent-person review.
`pr-corrections`	Commit-pinned PR snapshot	Focused factual review of corrected GPT-2 XL, AdamW, retrieval, loss, and leakage provenance.	pinned URL	The original GitHub review is mutable; primary paper, pinned upstream code, and canonical evidence remain stronger sources.
`run-positive-primary`	Run report	Narrative report for the positive-only primary row.	pinned URL	Concise narrative; exact outputs and scores defer to its hash-bound JSON.
`run-positive-conservative`	Run report	Narrative report for the positive-only conservative row.	pinned URL	Concise narrative; exact outputs and scores defer to its hash-bound JSON.
`run-positive-expanded`	Run report	Narrative report for the interrupted positive-only row.	pinned URL	No tuned evaluation; the user's interruption reason is retrospective context.
`run-paper`	Run report	Narrative report for the paper-adaptation row.	pinned URL	Contains stale upstream-optimizer wording corrected by the canonical retrospective.
`run-semantic-standard`	Run report	Narrative report for the semantic-standard row.	pinned URL	Concise narrative; exact outputs and scores defer to its hash-bound JSON.
`run-semantic-gentle`	Run report	Narrative report for the semantic-gentle row.	pinned URL	Concise narrative; exact outputs and scores defer to its hash-bound JSON.
`run-minimal-primary`	Run report	Narrative report for the minimal-pair primary row.	pinned URL	Concise narrative; exact outputs and scores defer to its hash-bound JSON.
`run-minimal-conservative`	Run report	Narrative report for the minimal-pair conservative row.	pinned URL	Concise narrative; exact outputs and scores defer to its hash-bound JSON.
`run-minimal-expanded`	Run report	Narrative report for the minimal-pair expanded row.	pinned URL	Concise narrative; exact outputs and scores defer to its hash-bound JSON.
`eval-collection`	Exact evaluation collection	All eight manifest-bound evaluation JSON records and their shared recorded environment.	pinned URL	The manifest supplies the authoritative run/path/hash bindings; exact filenames are linked below.
`eval-positive-primary`	Exact evaluation JSON	Positive-only primary prompts, byte-exact generations, metrics, configuration, Trainer runtime, and hardware.	pinned URL	Does not embed the training run ID; binding depends on the manifest path and SHA-256.
`eval-positive-conservative`	Exact evaluation JSON	Positive-only conservative prompts, byte-exact generations, metrics, configuration, Trainer runtime, and hardware.	pinned URL	Does not embed the training run ID; binding depends on the manifest path and SHA-256.
`eval-paper`	Exact evaluation JSON	Paper-adaptation prompts, byte-exact generations, metrics, configuration, Trainer runtime, and hardware.	pinned URL	Its configured 64-token cap alone does not prove why one output ends abruptly.
`eval-semantic-standard`	Exact evaluation JSON	Semantic-standard prompts, byte-exact generations, metrics, configuration, Trainer runtime, and hardware.	pinned URL	Observational evidence; it does not establish causes of recall or control errors.
`eval-semantic-gentle`	Exact evaluation JSON	Semantic-gentle prompts, byte-exact generations, metrics, configuration, Trainer runtime, and hardware.	pinned URL	Observational evidence; it does not establish causes of recall or control errors.
`eval-minimal-primary`	Exact evaluation JSON	Minimal-pair primary prompts, byte-exact generations, metrics, configuration, Trainer runtime, and hardware.	pinned URL	Observational evidence; it does not establish causes of specificity or retention errors.

Identifier	Source class	Supported claim scope	Pinned or durable URL	Limitation
`eval-minimal-conservative`	Exact evaluation JSON	Minimal-pair conservative prompts, byte-exact generations, metrics, configuration, Trainer runtime, and hardware.	`pinned URL`	Observational evidence; it does not establish causes of specificity or retention errors.
`eval-minimal-expanded`	Exact evaluation JSON	Minimal-pair expanded prompts, byte-exact generations, metrics, configuration, Trainer runtime, and hardware.	`pinned URL`	Observational evidence; it does not establish causes of specificity or retention errors.
`attestation-log-audit`	Retrospective author attestation	August 4, 2026 local verification that all nine ignored operational-log byte streams each matched the corresponding manifest SHA-256.	`pinned URL`	The URL exposes expected digests only; public readers cannot inspect or independently hash the local bytes or paths.

The eight exact JSON files are paired with rendered Markdown files under the same evidence commit. The JSON is authoritative for quoted model text; the Markdown is a derived human-readable rendering, and the manifest binds both by path and SHA-256.[CS:manifest][CS:eval-collection]

Structured JSON files[CS:eval-collection]: positive-primary, positive-conservative, paper-adaptation, semantic-standard, semantic-gentle, minimal-primary, minimal-conservative, and minimal-expanded.

The eight rendered companions are the positive-primary, positive-conservative, paper-adaptation, semantic-standard, semantic-gentle, minimal-primary, minimal-conservative, and minimal-expanded Markdown files.[CS:manifest]

The nine concise attempt reports are the positive-primary, positive-conservative, interrupted-positive-expanded, paper-adaptation, semantic-standard, semantic-gentle, minimal-primary, minimal-conservative, and minimal-expanded. Each row above resolves to its corresponding report marker as well as the manifest, evaluation JSON when present, and historical implementation. [CS:manifest]

The historical paper-run report contains stale wording about the upstream single-edit optimizer boundary. The corrected retrospective and pinned review record establish the narrower statement used here: the upstream paper used GPT-2 XL and its released code passed full model parameters to AdamW, whereas this project used a Qwen language-only LoRA adaptation. The upstream paper specifies Sentence-BERT and 15 nearest facts, but the required source pool, checkpoint, retrieval assets, and executable construction were not identified in the pinned released tree; this project's 15 locality facts were checked-in relation-matched examples, not a claimed reproduction of those neighbors. [CS:retrospective][CS:pr-corrections]